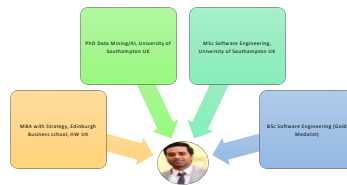


Part-I Recommender Systems

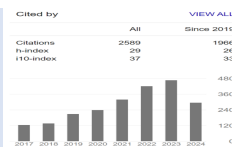
Theory and Implementation

Dr. Mustansar Ali Ghazanfar

1



- **Multi-disciplinary** experience: Associate professor, Project Director, AI Engineer, Data Team Lead, Assistant Manager
- **Spectrum:** Academia, Consultancy, Research & Development, Govt Sector, NGOs, Entrepreneurship
- **Prestige:** VC Award for Industry 4.0, 2024, OfS Funding 1M£, NHS/ Commonwealth collaborations, EU funding,



VIEW ALL

Prof. Adam Prugel Bennett
Head Machine Learning group, University of Southampton UK

Dr. Craig Saunders
Head, Applied Science, Amazon Alexa, UK

Prof. Nick Jennings
ViceProvost (Research and Enterprise)
Imperial College London

Sir Tim Berners-Lee
inventor www

Associate professor, UEL
Data Science, Artificial Intelligence, Strategy, Leadership, Innovation, Decision making and business optimization & Reforms, Automation, Personalisation, Industry 4.0

2

Learning Outcomes

By the end of this lecture on Recommender Systems, you will be able to:

- Define key concepts in Recommender Systems.
- Differentiate Collaborative Filtering, Content-Based Filtering,
- Demographics-Based Systems, and Hybrid Systems.
- Dimensionality Reduction based
- Implement in pyspark (lab)

3

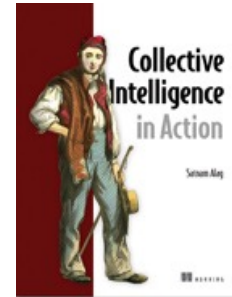
Agenda

- ❑ Introduction of Recommender Systems (RS)
- ❑ Types of Recommender Systems (Methodology used)
 - ❑ Collaborative Filtering systems
 - ❑ Content-based filtering systems
 - ❑ Demographic based systems
 - ❑ Hybrid Systems
- ❑ Dimensionality Reduction
- ❑ Conclusion

4

Collective Intelligence

- ❑ Wisdom of the crowd
- ❑ Some pattern emerges
- ❑ Some intelligence is found which was not visible at individual level.
- ❑ Many applications
 - ❑ Collaborative filtering
 - ❑ social media analysis
 - ❑ Stock market prediction
 - ❑ Tag cloud analysis



Satnam Alag

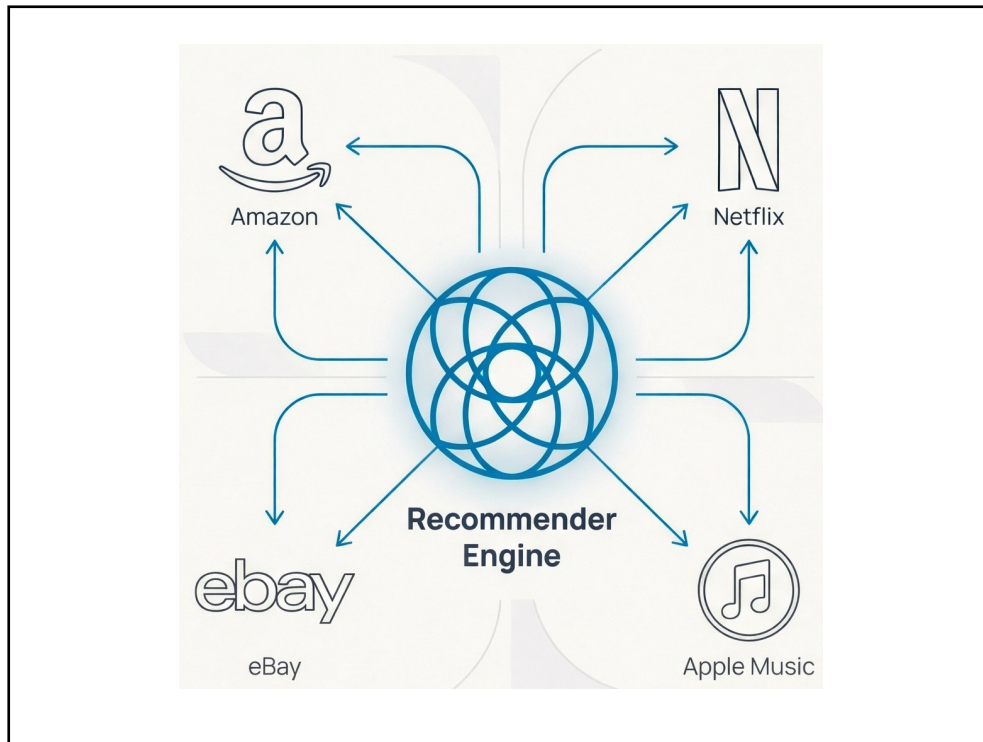
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Recommender Systems (RS)

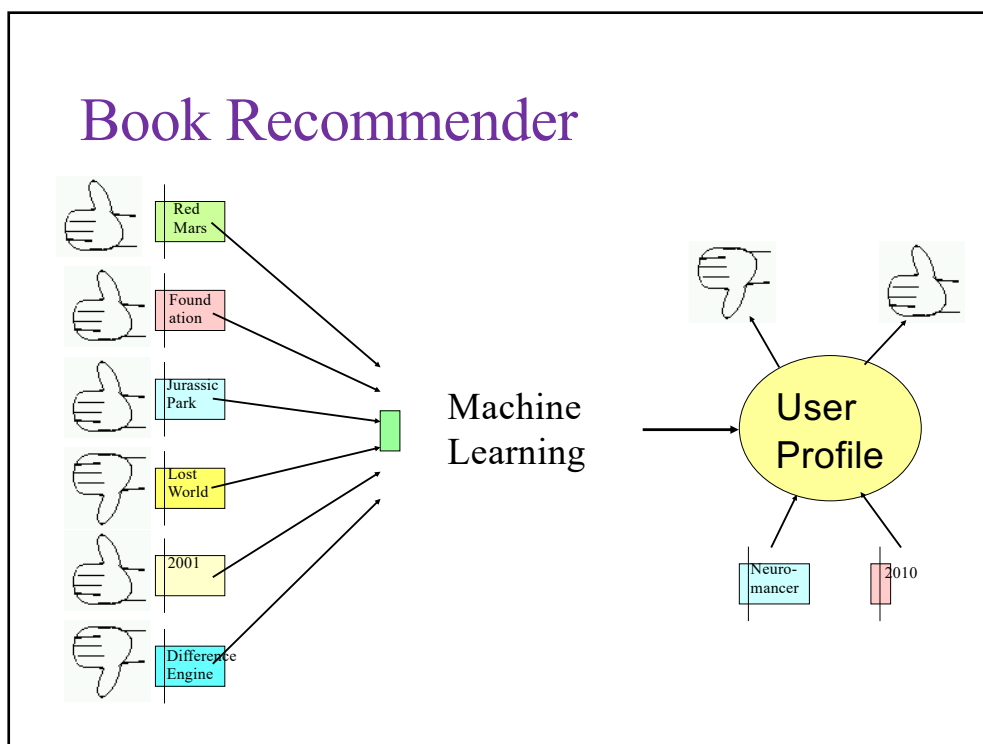
- ❑ Information filtering systems that make recommendations on items based on a model of user preferences.
- ❑ Examples



6



7



8

User and Item profiles

- ❑ **User profile** is made of ratings (about items) they provided
- ❑ **Item profile** is made of ratings given by other users, item's content features
- ❑ A data base of rating exists called **training set**
- ❑ User x item matrix called **user-item rating matrix**
 - ❑ very **sparse matrix**

9

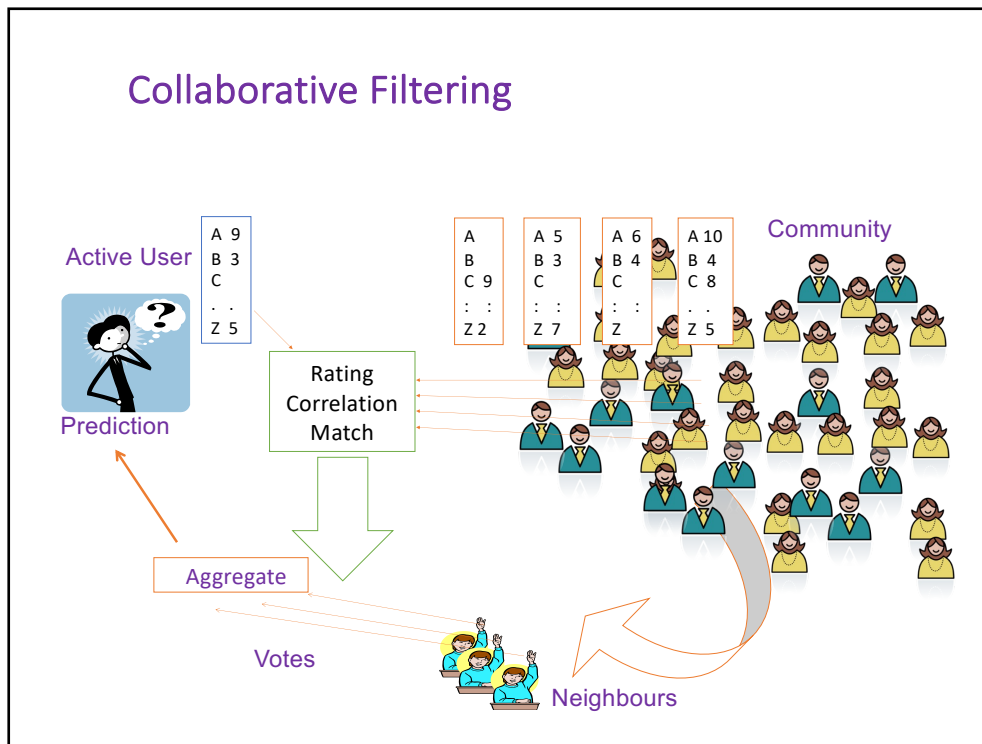
Collaborative Filtering

- ❑ Maintain a database of many users' ratings of a variety of items.
- ❑ For a given user, find other similar users whose ratings strongly correlate with the current user.
- ❑ Recommend items rated highly by these similar users, but not rated by the current user.
- ❑ **Almost all existing commercial recommenders use this approach** (e.g. Amazon).

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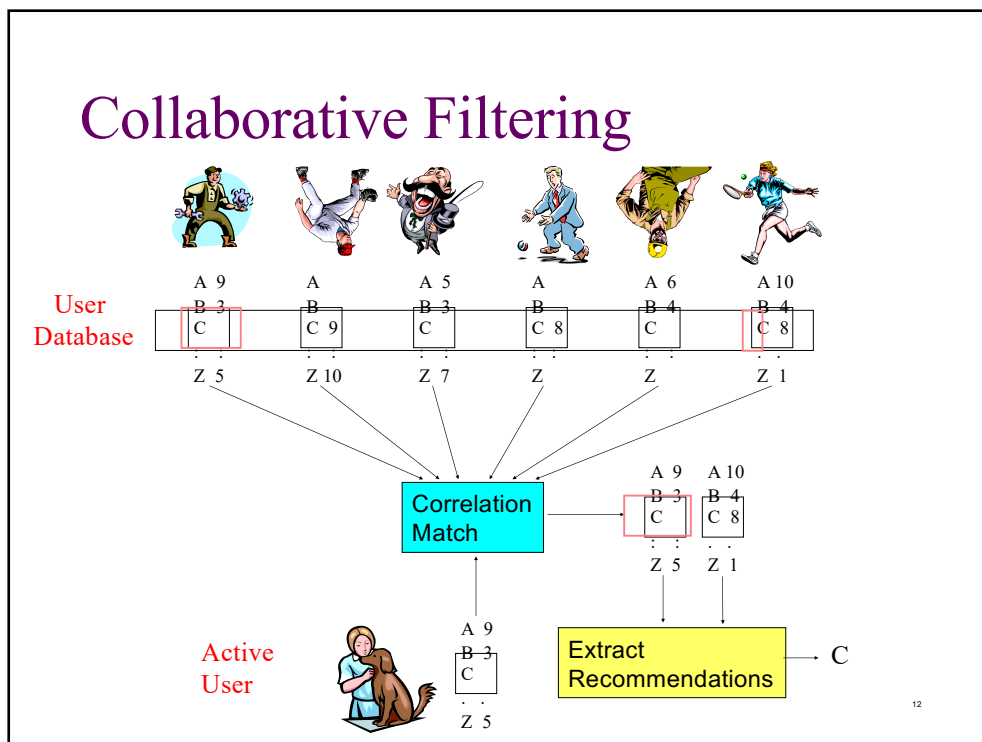
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Collaborative Filtering



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Collaborative Filtering



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User-Based CF

	Item1	Item2	Item3	Item4	Item5	Item6
User1		4	X	1	5	
User2	5	5	5	1		
User3	4	4	4		1	
User4	3		3		5	

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User-Based CF



	Item1	Item2	Item3	Item4	Item5	Item6
User1		4	X	1	5	
User2	5	5	5	1		
User3	4	4	4		1	
User4	3		3		5	

$$\text{Sim}(u1, u2) = \{-1, +1\} \rightarrow 0.5$$

$$\text{Sim}(u1, u3) = \{-1, +1\} \rightarrow -0.1$$

$$\text{Sim}(u1, u4) = \{-1, +1\} \rightarrow 0.3$$

$$\text{Sum}(|\text{sim}|) = 0.5 + 0.1 + 0.3 = 0.9$$

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User-Based CF

	Item1	Item2	Item3	Item4	Item5	Item6
User1		4	X	1	5	
User2	5	5	5	1		
User3	4	4	4		1	
User4	3		3		5	

$$\text{Prediction} = \{(\text{Sim}(u1,u2) * \text{rating_User2}) + (\text{Sim}(u1,u3) * \text{rating_User3}) + (\text{Sim}(u1,u4) * \text{rating_User4})\} / \text{Sum}(\text{sim})$$

$$\text{Prediction} = \{(0.5 * 5) + (-0.1 * 4) + (0.3 * 3)\} / (0.9) = 3.05$$

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User-Based CF

	Item1	Item2	Item3	Item4	Item5	Item6
User1		4	2	1	5	
User2	5	5	5	1		X
User3	4	4	4		1	
User4	3		3		5	

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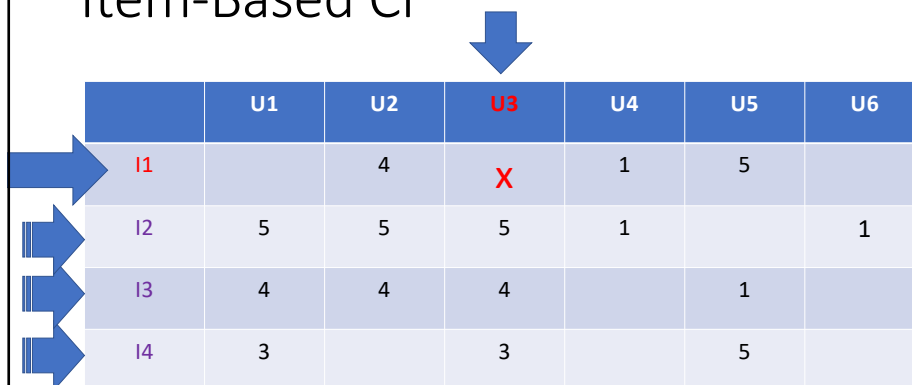
Item-Based CF

	Item1	Item2	Item3	Item4	Item5	Item6
User1		4	X	1	5	
User2	5	5	5	1		1
User3	4	4	4		1	
User4	3		3		5	

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Item-Based CF



	U1	U2	U3	U4	U5	U6
I1		4	X	1	5	
I2	5	5	5	1		1
I3	4	4	4		1	
I4	3		3		5	

$$\text{Sim}(I1, I2) = \{-1, +1\} \rightarrow 0.4$$

$$\text{Sim}(I1, I3) = \{-1, +1\} \rightarrow -0.1$$

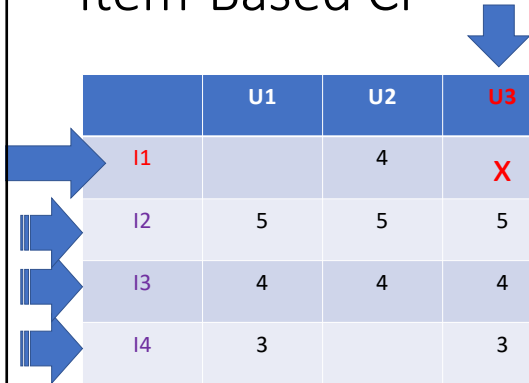
$$\text{Sim}(I1, I4) = \{-1, +1\} \rightarrow 0.3$$

$$\text{Sum}(|\text{sim}|) = 0.4 + 0.1 + 0.3 = 0.8$$

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Item-Based CF



	U1	U2	U3	U4	U5	U6
I1		4	X	1	5	
I2	5	5	5	1		1
I3	4	4	4		1	
I4	3		3		5	

$$\text{Prediction} = \{(\text{Sim}(I1, I2) * \text{rating_User3_I2}) + (\text{Sim}(I1, I3) * \text{rating_User3_I3}) + (\text{Sim}(I1, I4) * \text{rating_User3_I4})\} / \text{Sum}(\text{sim})$$

$$\text{Prediction} = \{(0.4 * 5) + (-0.1 * 4) + (0.3 * 3)\} / (0.9) = 2.85$$

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What is similarity?

- Find similar users, Similarity measures.

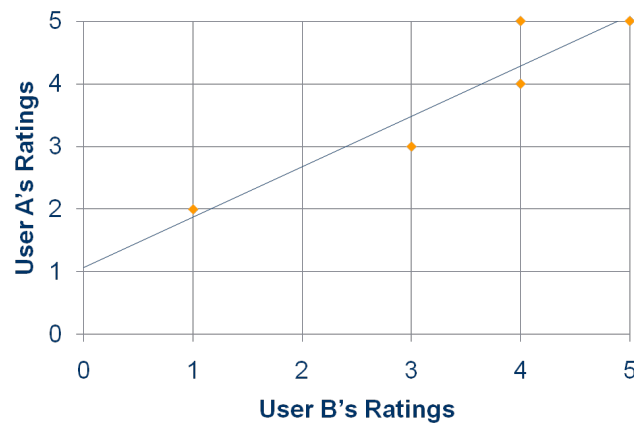
- Vector Similarity
$$w(a, i) = \sum_j \frac{v_{a,j}}{\sqrt{\sum_{k \in I_a} v_{a,k}^2}} \frac{v_{i,j}}{\sqrt{\sum_{k \in I_i} v_{i,k}^2}}$$

- Pearson Correlation

$$w(a, i) = \frac{\sum_j (v_{a,j} - \bar{v}_a)(v_{i,j} - \bar{v}_i)}{\sqrt{\sum_j (v_{a,j} - \bar{v}_a)^2} \sqrt{\sum_j (v_{i,j} - \bar{v}_i)^2}}$$

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Pearson Correlation



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Cold start scenarios

- ❑ New user cold start scenario, where the user is new and we don't have any data about them
- ❑ New item cold start scenario, where item is new and no one has rated it
- ❑ New system cold start scenario, where the system is new so the problem is how to bootstrap it?

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Types of Collaborative Filtering (CF)

- ❑ Memory based approaches
 - User-based CF
 - Item-based CF
- ❑ Model-based approaches
 - Clustering
 - MF
 - SVD etc

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Advantages with Collaborative Filtering

- Simple algorithm
- Use wisdom of crowd
- Can recommend 'out of the box'
- High quality recommendations (high accuracy)

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Problems with Collaborative Filtering

- **Cold Start:** There needs to be enough other users already in the system to find a match.
- **Sparsity:** If there are many items to be recommended, even if there are many users, the user/ratings matrix is sparse, and it is hard to find users that have rated the same items.
- **First Rater:** Cannot recommend an item that has not been previously rated.
 - New items
- **Popularity Bias:** Cannot recommend items to someone with unique tastes.
 - Gray-Sheep Users (i.e. outlier problem)

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Content-Based Recommending (CBR)

- Recommendations are based on information on the **content** of items rather than on other users' opinions.
- Uses a **machine learning algorithm** to induce a profile of the users preferences from examples based on a featural description of content.

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Advantages of CBR

- No cold-start or sparsity problems.
- Able to recommend to users with unique tastes.
- Able to recommend new and unpopular items
- Can provide explanations of recommended items by listing content-features that caused an item to be recommended.

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Disadvantages of CBR

- Requires content that can be encoded as meaningful features (problem for multimedia data).
- Users' tastes must be represented as a learnable function of these content features.
- Low quality recommendation (low accuracy)
- Unable to exploit quality judgments of other users.

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Content-Based Recommending Vs collaborating Filtering

Content Based Filtering	Collaborating Filtering
Info about content of items	Info about the similar users (or items)
Features & keywords about item	Ratings & Past history of user
Train Machine Learning classifiers and predict	Use (social) collective Intelligence to predict

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Feature Extraction for CBF

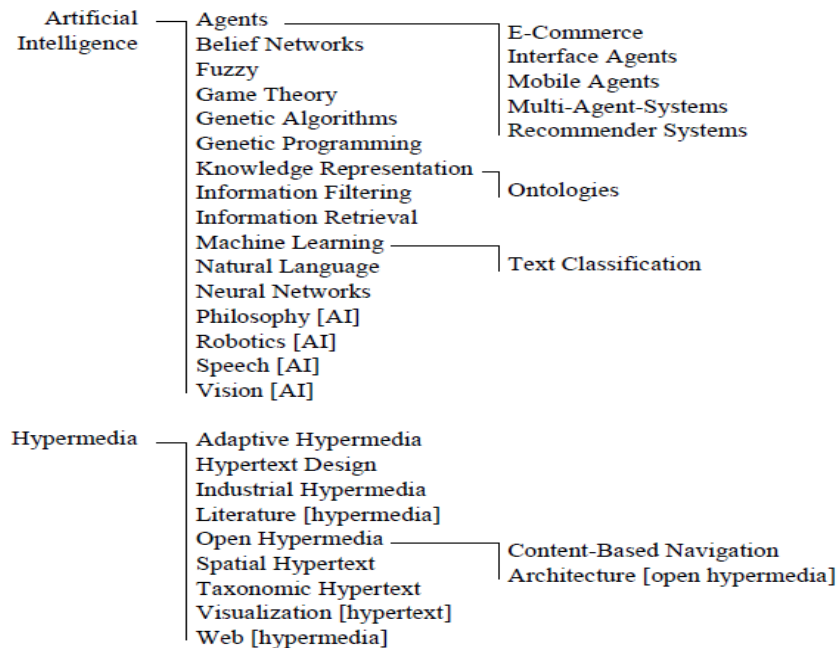
- ❑ Identify specific pieces of information in a unstructured or semi-structured textual document.
- ❑ Steps [9]
 - Tokenization
 - Stop word removal
 - Normalization
 - Stemming
 - Indexing
 - Weighting schemes

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Ontology-Based Recommendations in CBF

- Represent user and item profiles by ontology
- Ontology inferences (is-a, etc.)
- Benefits claim
 - Accuracy
 - Overcome some problems
 - Coverage

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Demographic-Based RS

- ❑ Cluster users/items based on demographic features
- ❑ Demographic features:
 - User
 - ✓ age, gender, post code, etc.
 - Item
 - ✓ Movie → horror, romantic, etc.

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34



- The video does
does have a Seat
- For instance, im
- In terms of psyc
- perhaps more br
- Although childre
- instance, initiatir

Marketing & Segmentation and Recommendations

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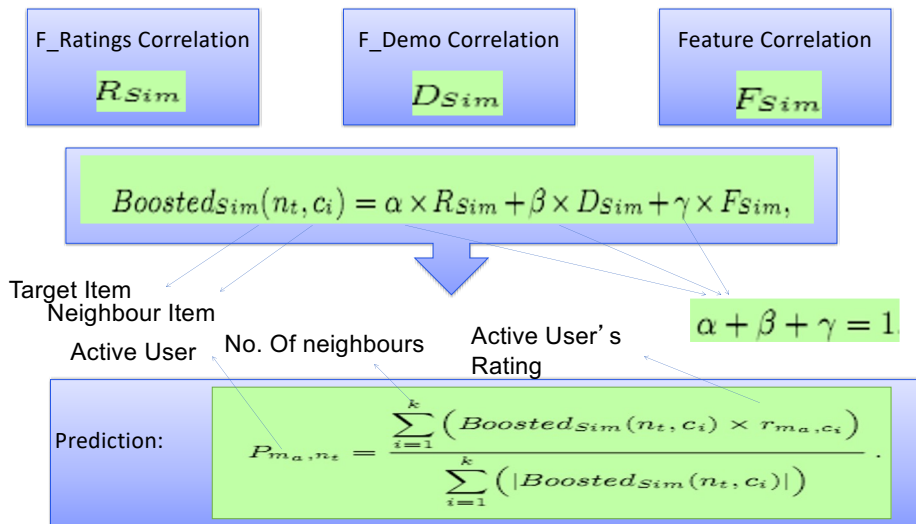
Segmentation



Single segment concentration—because the firm is concentrating marketing efforts to appeal to one particular segment – the luxury consumer.

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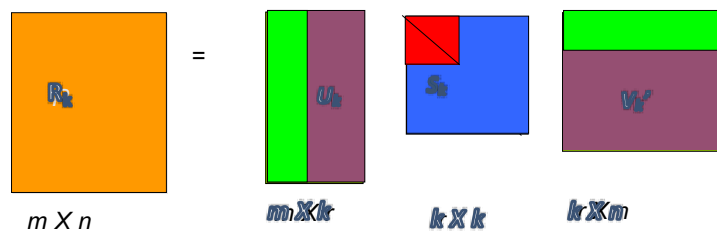
Integration Framework



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SVD-Based Recommendations

- ❑ Matrix factorization based technique
- ❑ SVD decomposes a matrix into three factors

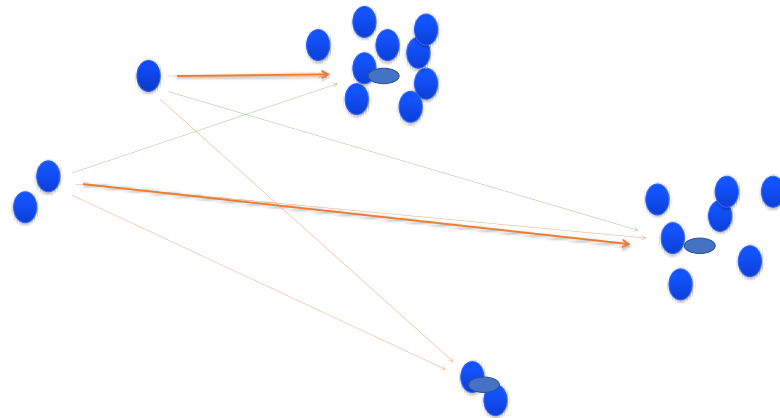


$$R_k = U_k \cdot S_k \cdot V_k' \approx R \text{ (K-Rank approximation)}$$

This reduced matrix can be used to predict ratings
Which were not present in the original matrix

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Clustering-Based Recommendations



Slow Convergence?
Cluster Sizes?

Centroid selection?
Online One pass

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Things you need to know

- ❑ Collective Intelligence
- ❑ Recommender systems
- ❑ Collaborative Filtering
- ❑ Content based filtering
- ❑ Demographic based systems
- ❑ Hybrid systems
- ❑ SVD
- ❑ Clustering

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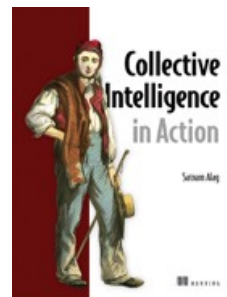
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That's all folk!

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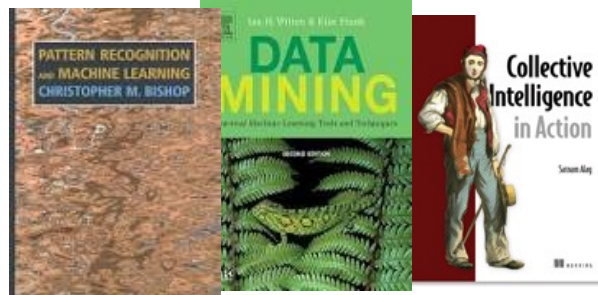
Online Resources

- Coursera
 - <https://www.coursera.org/course/recsys>
- Robust, scalable, and practical algorithms for recommender systems, *Mustansar Ali Ghazanfar*,
 - <http://eprints.soton.ac.uk/343761/>
- Toward the next generation of recommender systems: a survey of the state-of-the-art and possible extensions, *Adomavicius, G.*



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References and Online Resources

- LinkedIn: <https://www.linkedin.com/learning/comptia-data-plus-da0-001-cert-prep-domain-3-0-data-analysis/data-analysis?u=56744289>
- Interview Tips: <https://www.linkedin.com/advice/1/what-best-ways-prepare-data-analysis-interviews>